

Structure-Aware Multimodal Learning Improves Minor-Class Signal Peptide Prediction

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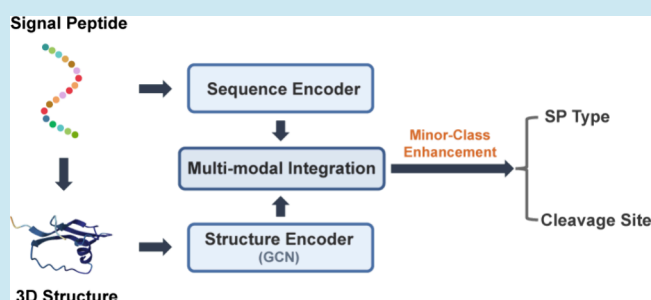
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ABSTRACT: Signal peptides play essential roles in protein secretion and localization, and their accurate identification is critical for understanding protein synthesis, transport, and functional regulation. However, severe class imbalance in signal peptide data sets leads to substantially lower recognition performance for minor classes compared with major classes. Here, we propose a structure-aware multimodal signal peptide prediction network (SaSPNet), which incorporates structural modality information into conventional sequence modeling and uses a graph convolutional network (GCN)-based structure encoder to learn structural representations of signal peptides for both signal peptide type and cleavage-site prediction. SaSPNet significantly improves the prediction performance for minor signal peptide classes on the USPNet data set, achieving more than a 10% gain over existing methods on key minor-class metrics. Feature visualization and explainability analyses show that the structure encoder learns more discriminative structural patterns for minor signal peptides, revealing the mechanism by which the structural modality enhances model performance. In addition, comparative analyses using three-dimensional structures generated by different structure prediction models demonstrate that SaSPNet is robust to variations in structural data quality. We further construct an independent test set, SP-MinorEval, specifically for minor signal peptides, and evaluations on this data set show that SaSPNet maintains strong performance across domains, providing an effective tool for minor-class signal peptide prediction, protein secretion mechanism studies, and functional protein discovery.

KEYWORDS: signal peptide prediction, minor-class signal peptide, protein structure modeling, graph convolutional network, multimodal integration, structure-aware learning



1. INTRODUCTION

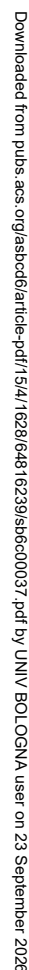
Signal peptides (SPs) are short amino-terminal sequences typically composed of a positively charged N-region, a hydrophobic H-region, and an uncharged C-region.¹ They direct newly synthesized proteins into the secretory pathway and are recognized and cleaved at a site near the C-region upon arrival at their destination, playing a key role in protein secretion and transport.² Given their importance, accurate identification of signal peptides and their cleavage sites is essential for understanding protein synthesis, secretion, and biological function in depth.³

Because experimental identification of signal peptides is often time-consuming and costly,^{4,5} researchers have developed various computational tools to predict SP types and cleavage sites. Early studies trace back to von Heijne's empirically derived statistical rules for identifying cleavage sites; although pioneering, this approach relied on limited data and therefore achieved modest accuracy.⁶ Subsequent homology-based methods emerged, such as Signal-BLAST proposed by Frank and Sippl, which determines whether an input sequence contains a signal peptide by aligning it against

known sequences in a database, enabling binary prediction but showing limited generalization, particularly for novel species or low-homology proteins.⁷

With the accumulation of data and advances in modeling techniques, machine learning methods have been increasingly applied to SP prediction. SignalP1.0, proposed by Nielsen et al., employed neural networks for binary classification and cleavage-site prediction, achieving substantially higher accuracy than statistical approaches but frequently misclassifying signal anchors.⁸ Subsequent versions, SignalP2.0 and SignalP3.0, combined neural networks with hidden Markov models and introduced the D-score, expanding prediction capability and attempting to characterize different SP regions; however, their generalization to signal anchors and cleavage-site accuracy

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Almagro Armenteros et al., combined CNNs, bidirectional long short-term memory networks (BiLSTMs),¹⁵ and conditional random fields (CRFs)¹⁶ to predict four SP types, though it did not cover all SP categories. SignalP6.0,¹⁷ developed by Teufel et al., leveraged the protein language model ProtBERT¹⁸ to achieve six-class prediction for five SP types plus non-SPs, and further improved cleavage-site prediction, yet some classes still suffered from limited generalization. USPNet,¹⁹ proposed by Shen et al., integrated a protein

language model with a BiLSTM-attention module and employed the LDAM loss²⁰ to mitigate class imbalance, achieving superior overall performance and eliminating reliance on species labels. However, its predictions still relied solely on sequence information and performed poorly for certain minor-class SPs. Although recent protein language model-based multitask frameworks have demonstrated strong sequence representation capabilities in complex tasks such as protein–peptide interactions,²¹ reliance solely on sequence modality remains insufficient for predicting minor-class SPs.

Overall, these methods have achieved significant success in the overall performance of SP prediction; however, due to severe class imbalance in SP data sets, the prediction performance for minor-class SPs (e.g., Sec/SPII, Sec/SPIII, Tat/SPI, Tat/SPII) still lags significantly behind that of major classes.¹⁹ Although samples of these minor-class SPs are scarce, they play important roles in biological studies such as protein secretion mechanisms and the identification of pathogenic factors.^{22–24} Therefore, improving the prediction of minor classes while maintaining overall performance remains a critical challenge in the field of SP prediction.

Recent advances in protein structure prediction models have made it possible to obtain high-quality protein three-dimensional structures at scale,²⁵ providing an opportunity to incorporate structural information into SP prediction. Inspired by this, we developed a structure-aware multimodal SP prediction model (SaSPNet), which integrates structural features on top of sequence feature modeling to simultaneously predict SP classes and cleavage sites. Specifically, we employ a CNN-BiLSTM-attention module as the sequence encoder to capture SP sequence patterns and use the protein language model ESM-2²⁶ to supplement global semantics and evolutionary information. Meanwhile, a graph convolutional network (GCN)²⁷ serves as the structure encoder, learning the topological relationships between residues from 3D structures predicted by tools such as ESMFold and AlphaFold2.^{25,26} By fusing sequence and structural features, SaSPNet maintains overall SP prediction performance while significantly improving the prediction of minor-class SPs (Figure 1).

2. METHODS

2.1. Data Set

2.1.1. Training and Benchmark Data Sets. We used the same training and benchmark data sets as USPNet¹⁹ to ensure comparability of results. This data set is primarily derived from the data used by SignalP6.0¹⁷ and has been further extended and processed: additional signal peptide samples were incorporated from the UniProt²⁸ and Prosite²⁹ databases; soluble and transmembrane proteins were added from UniProt and TOPDB³⁰; and a homology partitioning strategy³¹ was applied to limit sequence similarity to below 30%. The final training set contains 13,679 sequences, and the benchmark set contains 6611 sequences.

This data set covers five types of signal peptides (Sec/SPI, Sec/SPII, Sec/SPIII, Tat/SPI, Tat/SPII) as well as nonsignal peptides (NO-SP), forming a six-class classification task. As shown in Table 1, the number of samples varies significantly across classes, with Sec/SPI (SP) and NO-SP accounting for the major, while Sec/SPII (LIPO), Sec/SPIII (PILIN), Tat/SPI (TAT), and Tat/SPII (TATLIPO) represent minor classes, resulting in a typical long-tail distribution.

2.1.2. Acquisition and Processing of 3D Structures. Based on the training and benchmark sets, we further constructed three-dimensional structure data for signal peptides. Specifically, we used advanced protein structure prediction models such as ESMFold²⁶ to

Table 1. Statistics for the Dataset

class	training set	test set	total	proportion (%)	notes
NO-SP	9749	5876	15,625	77.00	major-class
Sec/SPI	2324	258	2582	12.73	major-class
Sec/SPII	1229	386	1615	7.96	minor-class
Tat/SPI	287	78	365	1.80	minor-class
Tat/SPII	20	13	33	0.16	minor-class
Sec/SPIII	70	0	70	0.34	minor-class
Total	13,679	6611	20,290		

generate corresponding 3D structure PDB files from the input signal peptide sequences.

To incorporate 3D structural features into the model, we represented the signal peptide 3D structures as graph structures, with amino acid residues as nodes and spatial adjacency relationships between residues as edges. Specifically, we extracted the 3D coordinates of residues from each sample's PDB file and set a spatial distance threshold of 10 Å; two residues were considered adjacent if the distance between their C α atoms was below this threshold.

2.1.3. Independent Test Set. To evaluate SaSPNet's domain-shifted generalization ability on minor-class signal peptides and to address the absence of Sec/SPIII (PILIN) samples in the USPNet benchmark test set, we constructed an independent test set named SP-MinorEval. This data set focuses on four minor-class signal peptides: Sec/SPII (LIPO), Sec/SPIII (PILIN), Tat/SPI (TAT), and Tat/SPII (TATLIPO), containing a total of 792 signal peptide sequences. The specific class distribution is shown in Figure 4f.

The construction process of SP-MinorEval is as follows: (1) Extract minor-class samples from the SignalP6.0 test set and USPNet test set, and take the union of both as the original minor-class collection; (2) Perform sample deduplication within this collection to eliminate duplicate sequences; (3) Remove samples that overlap with the USPNet training set, as well as samples that overlap with the USPNet test set in the Sec/SPII and Tat/SPI classes; (4) Use CD-HIT³² to reduce the homology within the collection to below 90%.

2.2. Model Architecture

The input data of SaSPNet consist of the amino acid sequence of a signal peptide and its corresponding residue contact graph derived from the predicted three-dimensional structure. Considering that signal peptides are typically short, the maximum sequence length is set to 70 residues. Given these inputs, SaSPNet follows a unified workflow in which sequence-based and structure-based information are first encoded in parallel. Specifically, the signal peptide sequence is processed in parallel by a sequence encoder and a protein language model to capture contextual and evolutionary features, while the residue contact graph is modeled by a structure encoder based on graph convolutional networks to extract structural representations. The resulting multimodal features are then integrated to form the integrated features, which are subsequently fed into two task-specific prediction heads for signal peptide type classification and cleavage site prediction.

2.2.1. Sequence Encoder. The amino acid residue sequence is first fed into an embedding layer to obtain initial sequence embeddings for subsequent processing. The embedding vectors are then used in two branches, entering the CNN block and the BiLSTM-attention block, respectively, to extract features from both local patterns and long-range dependencies.

In the CNN block, we employ a one-dimensional convolutional network to learn key motifs within the signal peptide sequence. The convolution aggregates the semantics of neighboring residues within a sliding window, yielding features H_{CNN} that represent local patterns of the signal peptide.

In the BiLSTM-attention block, the embedding vectors are first fed into a BiLSTM layer. The BiLSTM models directional information from the N-terminus to the C-terminus and from the C-terminus to the N-terminus simultaneously, generating a bidirectional contextual

representation. For position t , the bidirectional state can be expressed as

$$\mathbf{h}_t = [\vec{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] \quad (1)$$

where $\vec{\mathbf{h}}_t$ and $\overleftarrow{\mathbf{h}}_t$ represent the forward and backward hidden states of the BiLSTM at position t , respectively.

The BiLSTM output is then processed by multihead attention³³ for feature aggregation, allowing the model to focus on distal residues that influence the current position. Each attention head uses scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

where Q , K , and V are the query, key, and value matrices, respectively, d_k is the dimension of the query and key, and T denotes the matrix transpose operation.

The output of each attention head can be expressed as

$$\text{head}_i = \text{Attention}(QW_Q^i, KW_K^i, VW_V^i) \quad (3)$$

where W_Q^i , W_K^i , and W_V^i are the parameter matrices, and i denotes the i th attention head. Multiple attention heads operate in parallel, and the resulting outputs are concatenated to produce the final multihead attention output:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_O \quad (4)$$

where W_O is the output projection matrix, and h denotes the number of attention heads.

The outputs of the multihead attention and the BiLSTM layer are concatenated to obtain the feature $\mathbf{H}_{\text{BiLSTM}}$ representing the global dependencies of the signal peptide. To integrate the local and global feature sources, we perform element-wise summation of the CNN block and BiLSTM–attention block outputs to obtain the sequence encoder feature:

$$\mathbf{H}_{\text{seq}} = \mathbf{H}_{\text{CNN}} + \mathbf{H}_{\text{BiLSTM}} \quad (5)$$

2.2.2. Protein Language Model. Protein language models can learn latent evolutionary features solely from amino acid sequences, providing more expressive representations for downstream tasks. To further enhance the discrimination of different types of signal peptides, we incorporated the pretrained model ESM-2²⁶ as a global semantic supplement for sequences. In SaSPNet, each signal peptide sequence is input into the pretrained ESM-2 model to obtain its final-layer hidden representation. The resulting features are then pooled along the sequence dimension to generate sample-level protein language model features.

2.2.3. Structure Encoder. To explicitly integrate structural information into the model representation, SaSPNet employs a structure encoder based on Graph Convolutional Networks (GCN)²⁷ to extract signal peptides' structural topological features.

In the data processing stage, we construct the adjacency matrix $A \in \mathbb{R}^{L \times L}$ for the signal peptide residue contact graph, where $A_{ij} = 1$ indicates that the i th and j th residues are in spatial contact, and $A_{ij} = 0$ otherwise. We then add the identity matrix to the adjacency matrix to obtain the self-connected structural graph, and perform symmetric normalization on the adjacency matrix to obtain $\tilde{A}$.

GCN aggregates information from neighboring nodes based on the node features. For the l th layer of GCN, the input is the node features from the previous layer $H^{(l)} \in \mathbb{R}^{L \times C_l}$, and the output is the node representation $H^{(l+1)}$:

$$H^{(l+1)} = \text{ReLU}(\tilde{A}H^{(l)}W^{(l)}) \quad (6)$$

where $W^{(l)} \in \mathbb{R}^{C_l \times C_{l+1}}$ is the trainable weight matrix. In SaSPNet, a stack of three GCN layers is used to construct the structure encoder. The input node features of the first layer are the sequence encoder features, and the second and third layers take the output from the previous layer as their input. To fully preserve topological features

from different depths, we concatenate the outputs of the three GCN layers to obtain the final structural features:

$$H_{\text{struc}} = \text{Concat}(H^{(1)}, H^{(2)}, H^{(3)}) \quad (7)$$

2.2.4. Multimodal Fusion and Prediction Heads. To integrate heterogeneous information from different modalities, we adopt a feature-level fusion strategy based on direct concatenation. This design aims to preserve modality-specific representations while enabling complementary information from sequence-derived and structure-derived features to be jointly exploited. Compared with early fusion strategies that mix raw inputs or late fusion schemes that combine independent predictions, feature concatenation provides a simple yet effective way to maintain explicit feature representations from each modality and allows downstream prediction heads to learn task-specific combinations in a flexible manner.

Alternative multimodal fusion strategies, such as weighted summation, attention-based fusion, gating mechanisms, or bilinear pooling, have been explored in previous studies. While these approaches can model complex cross-modal interactions, they typically introduce additional parameters and increase optimization complexity, which may lead to overfitting under limited training data or severe class imbalance. In contrast, the concatenation-based fusion adopted in SaSPNet avoids premature information mixing and retains clear modality boundaries, which is particularly beneficial for minor-class signal peptide prediction, where stable and discriminative representations are crucial.

For the signal peptide classification task, following the above multimodal fusion strategy, we first concatenate the residue-level features from the sequence encoder and the structure encoder along the feature dimension. Then, we perform global pooling to obtain sample-level features. These are subsequently concatenated with the protein language model features from ESM-2, resulting in the final multimodal integrated features, which can be represented as

$$h_{\text{integrated}} = \text{Concat}(\text{Pool}(H_{\text{seq+struc}}), h_{\text{plm}}) \quad (8)$$

where $H_{\text{seq+struc}}$ represents the residue-level concatenated features from the sequence and structure encoders, $\text{Pool}(\cdot)$ denotes the global pooling operation, and h_{plm} represents the sample-level feature from ESM-2.

The integrated features are then fed into a prediction head implemented with a three-layer MLP, which outputs 6-dimensional logits corresponding to the 5 types of signal peptides and nonsignal peptides.

For the signal peptide cleavage-site prediction task, we output the classes at the residue level, so no global pooling is performed. Instead, we directly use the residue-level fused representation obtained by concatenating the sequence encoder features and the structure encoder features. The fused features are then input into another three-layer MLP prediction head, which outputs 11 classes of logits for each position in the sequence. The cleavage-site labeling system includes 10 valid residue labels, with an additional empty label added to padded positions for sequences shorter than 70 residues. Therefore, there are a total of 11 classes. The cleavage sites in the sequence can be determined from the 11 classes of logits produced by the model.

2.3. Loss Function

Signal peptide data sets generally exhibit significant class imbalance, with minor class samples being far fewer than nonsignal peptides or common types. This can cause models to be biased toward major classes during training, resulting in poor generalization to rare types. To address this, SaSPNet employs the Label-Distribution-Aware Margin (LDAM) loss function,²⁰ which adjusts class-dependent margins to enhance discrimination for minor classes.

The core idea of the LDAM loss function is to introduce a class-dependent margin into the softmax cross-entropy, allowing the decision boundaries of major and minor classes to shrink by different amounts. For class j , the margin is defined as

$$\Delta_j = \frac{C}{n_j^{1/4}} \quad (9)$$

where n_j denotes the number of samples in class j , and C is a tunable hyperparameter controlling the overall margin scale. In this work, C is not manually tuned but is determined according to the class distribution as $C = 0.5/\max_{1 \leq j \leq K}(n_j^{-1/4})$, which ensures that the maximum margin across all classes is bounded. For a sample with true label y , the corresponding logit z_y will be reduced by Δ_y . To avoid excessively large or small margins affecting the optimization, we introduce a scaling factor s to adjust the range of the normalized logits, which is set to $s = 1$ in our experiments. The final LDAM loss for a single sample is given by

$$\mathcal{L}(x, y) = -\log \left(\frac{e^{\frac{(z_y - \Delta_y)}{s}}}{e^{\frac{(z_y - \Delta_y)}{s}} + \sum_{i \neq y} e^{\frac{z_i}{s}}} \right) \quad (10)$$

In SaSPNet, the LDAM loss function is applied to both tasks: signal peptide type and cleavage-site prediction. The former performs classification at the sample level with $K = 6$ classes, while the latter performs prediction at the residue level with $K' = 11$ residue categories. The loss functions for these tasks are as follows:

$$\mathcal{L}_{\text{type}} = -\frac{1}{N} \sum_{j=1}^N \sum_{y=1}^K \mathcal{L}(x_j, y_j) \quad (11)$$

$$\mathcal{L}_{\text{cleavage-site}} = -\frac{1}{N \times L} \sum_{j=1}^{N \times L} \sum_{y=1}^{K'} \mathcal{L}(x_j, y_j) \quad (12)$$

The final optimization objective is

$$\mathcal{L}_{\text{SaSPNet}} = \mathcal{L}_{\text{type}} + \tau \mathcal{L}_{\text{cleavage-site}} \quad (13)$$

where N is the number of samples, $L = 70$ is the maximum sequence length, and τ is the weight coefficient for the two tasks. In our experiments, τ is set to 1, assigning equal importance to signal peptide type prediction and cleavage-site prediction.

2.4. Experimental Setup

All experiments in this study were conducted on an NVIDIA RTX 3090 GPU (24GB VRAM), running on an Ubuntu operating system with Python version 3.9.16. The deep learning framework used was PyTorch 1.13.1. Model training was performed using the Adam optimizer with an initial learning rate of 5×10^{-4} and weight decay set to 1×10^{-4} . To select the optimal hyperparameters, 5-fold cross-validation was performed, and the final configuration was determined based on the average performance across the five validation runs. The model was then formally trained on the full training set, with a maximum of 300 epochs, and early stopping was applied to prevent overfitting.

2.5. Performance Evaluation

To systematically evaluate SaSPNet's performance in signal peptide prediction, we assessed it on two task levels: signal peptide classification and cleavage-site prediction. Within each task, we further examined both overall performance and performance on minor classes.

2.5.1. Evaluation of Signal Peptide Classification Performance. In the signal peptide classification task, we first focus on the overall performance of the model across all six classes. To avoid potential bias in accuracy (ACC) under imbalanced data conditions, we use the multiclass Matthews Correlation Coefficient (MCC) as the evaluation metric, which considers all elements of the confusion matrix (TP, TN, FP, and FN) and is widely regarded as a robust metric for imbalanced classification, providing a more balanced and informative assessment than ACC. For the binary classification case, its formula is

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (14)$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. The multiclass MCC is a generalization of this metric to the multiclass confusion matrix.

In addition to overall performance, we further evaluate the performance on the four minor classes, specifically using the three metrics: Precision, Recall, and F1-score:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (15)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

F1-score is the harmonic mean of Precision and Recall, balancing false positives and missed detections, thereby providing a comprehensive assessment of the model's ability to identify each minor class.

In addition, to assess the model's ability to distinguish each individual minor class, we further adopt the One-vs-Rest (OvR) strategy to calculate class-level MCC. The target minor class is treated as the positive class, while the remaining five classes are treated as the negative class, and the MCC is computed using the formula mentioned above.

2.5.2. Evaluation of Signal Peptide Cleavage-Site Prediction Performance. For the cleavage-site prediction task, we also use Precision, Recall, and F1-score as the primary evaluation metrics. In this task, each sequence position may correspond to a cleavage site or a noncleavage site, and the metrics are defined consistently with the classification task, as shown in formulas 15 and 16. Specifically, predicted cleavage-site positions are compared with the ground-truth cleavage-site annotations at the residue level. A prediction is considered correct only when the predicted cleavage site exactly matches the annotated position, without allowing positional deviations. Precision is defined as the proportion of correctly predicted cleavage sites among all predicted cleavage sites, while Recall is defined as the proportion of correctly predicted cleavage sites among all ground-truth cleavage sites. These metrics are calculated separately for all six classes and for the four minor classes to assess the model's ability to localize cleavage sites in rare signal peptide types.

2.6. Explainability Analysis

In order to further understand the role of the structural modality in predicting minor class signal peptides, we conduct an explainability analysis of the structure encoder in SaSPNet. Specifically, we employ GNNExplainer³⁴ to explain the decision-making process of the GCN module. The core idea of GNNExplainer is to learn a set of trainable masks that select key edges and features from the input graph, which most strongly support the model's predictions. This leads to a sparse and high-confidence subgraph, revealing the residue topological connection patterns that the model focuses on.

We run GNNExplainer on each class of signal peptide samples individually to generate importance scores for the edges in the residue contact graph of each sample. From these, we select a fixed number of top-ranked key edges to construct a subgraph, thereby obtaining the critical residue contact subgraph for each sample in the structural modality. The fixed number of key edges ensures the comparability of the critical subgraphs across different samples, preventing the results from being affected by large variations in the number of edges in the residue contact graphs of different samples.

Next, we compare the within-class similarity of the critical subgraphs across different signal peptide classes to assess whether the GCN has learned stable and consistent structural patterns within the same class.

To calculate the within-class similarity, we first need to measure the similarity between the critical subgraphs of different samples. We use Jaccard similarity³⁵ and Cosine similarity³⁶ to quantify the similarity between subgraphs.

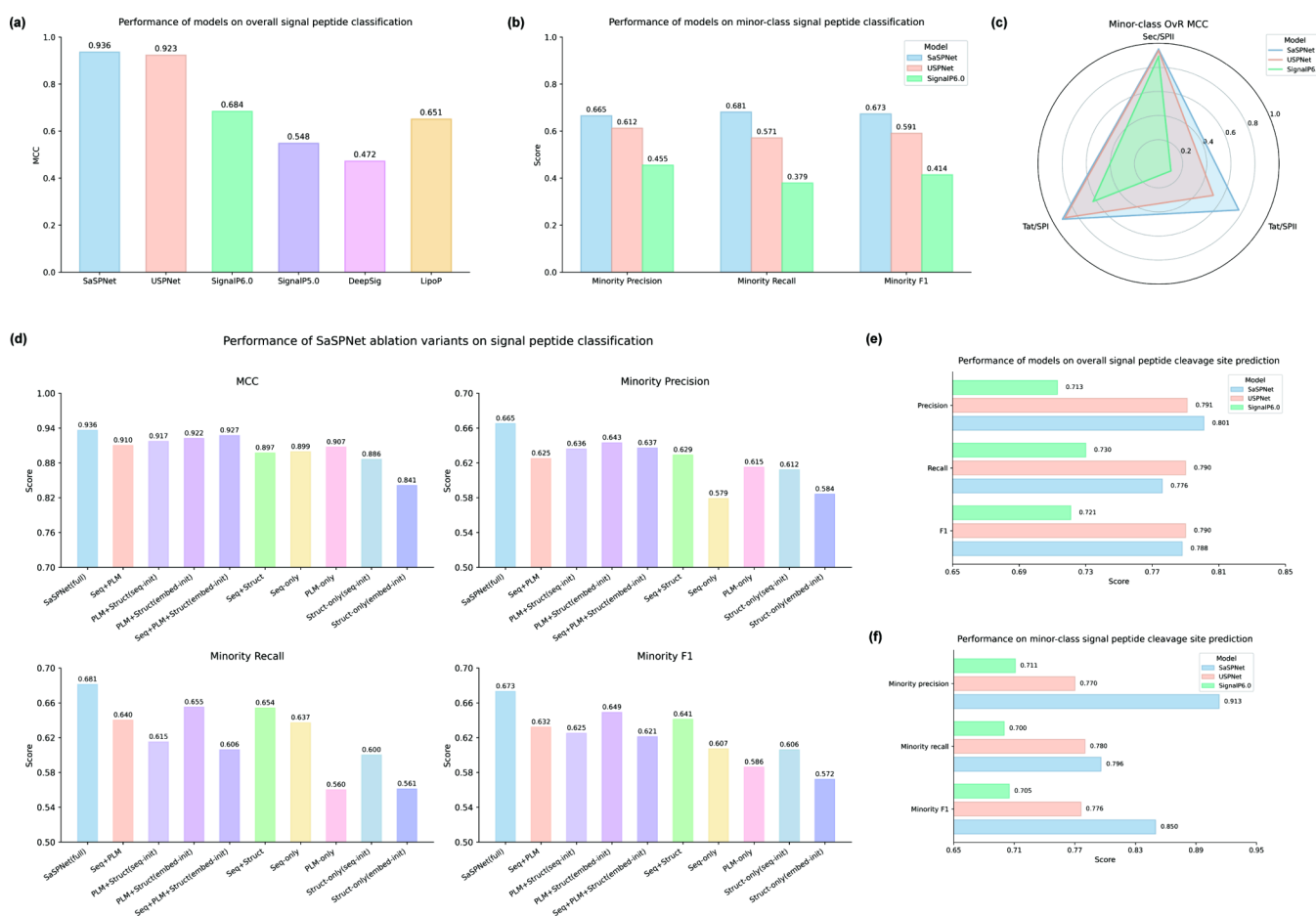


Figure 2. Performance comparison, ablation analysis, and cleavage-site prediction of SaSPNet. (a) Overall multiclass MCC on six SP types compared with state-of-the-art methods. (b) Performance of SaSPNet and baseline models on minor-class SP classification. We use Precision, Recall, and F1-score to measure minor-class performance, which correspond to Minority Precision, Minority Recall, and Minority F1 in the figure. (c) Radar chart of minor-class performance across models. To measure the ability to recognize individual minor-class SP types, we compute One-vs-Rest MCC by treating each minor-class against all other classes, denoted as OvR MCC in the figure. (d) Comparison of SP classification performance among different ablation variants of SaSPNet, including overall MCC and minor-class metrics. The horizontal axis lists the ablated model variants. *SaSPNet (full)* denotes the complete model. *Seq* indicates the sequence encoder, *PLM* refers to protein language model, and *Struct* corresponds to the structure encoder. *Struct (seq-init)* refers to the GCN block whose initial node features are taken from the sequence encoder output, while *Struct (embed-init)* uses the output of the embedding layer as the initial GCN node features. (e,f) Comparison of cleavage-site prediction performance across models, evaluated separately for overall SPs and minor-class SPs.

For any two samples' key edge sets A and B , the Jaccard similarity is defined as

$$\text{Jaccard}(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (17)$$

The critical subgraphs of the two samples are converted into fixed-size adjacency matrices, which are then flattened into vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, allowing the calculation of the Cosine similarity:

$$\text{Cosine}(\mathbf{v}_1, \mathbf{v}_2) = \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \|\mathbf{v}_2\|} \quad (18)$$

Based on this, for each class, we calculate the pairwise similarities between every two samples within the class and then take the arithmetic average of these similarities to obtain the within-class similarity. Additionally, we calculate the between-class similarities between different classes, which is done by computing the pairwise similarities between each sample pair from the two classes and taking the arithmetic average. The resulting within-class and between-class similarities form the similarity matrix.

To assess whether the within-class similarity of the minor classes differs significantly from that of the major class samples, we perform a t test³⁷ for statistical evaluation. The purpose of this test is to examine

whether the model captures more consistent structural patterns for minor classes compared to major classes. Specifically, we combine NO-SP and Sec/SPI(SP) into the major class group, and Sec/SPII(LIPO), Tat/SPI(TAT), and Tat/SPII(TATLIPO) into the minor class group.

Importantly, the statistical samples used in the t test are not the averaged within-class similarities of each class, but the pairwise similarity values computed between all sample pairs within each group. That is, all pairwise critical subgraph similarities obtained during the within-class similarity calculation are treated as individual observations. This results in a sufficiently large number of samples for each group, making the t test statistically applicable.

Under this formulation, the t test evaluates whether the distribution of within-class pairwise similarities in the minor class group is significantly higher than that in the major-class group. The t test statistic is defined as follows:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (19)$$

where $\bar{x}_1$ and $\bar{x}_2$ represent the average mean values of the pairwise within-class critical subgraph similarities for the major-class group and

the minor-class group, respectively, s_1^2 and s_2^2 are the variances of the two groups, and n_1 and n_2 represent the sample sizes of the two groups. Based on the calculated t -value and degrees of freedom, the corresponding significance level p can be obtained from the t -distribution table. If $p < 0.05$, it is considered that there is a significant difference in the within-class similarity between the two groups.

3. RESULTS

3.1. Performance Evaluation on SP Classification

We trained and evaluated SaSPNet using the USPNet training and benchmark test sets. For the overall signal peptide classification performance evaluation, we used MCC as the primary evaluation metric. To provide a comprehensive comparison of model performance, we selected five methods to compare with SaSPNet. Among them, USPNet and SignalP6.0 were trained and evaluated on the same training and test sets, while the test results of other methods were obtained from their respective web servers. The results show that SaSPNet significantly outperforms other methods in the overall signal peptide classification task and slightly exceeds USPNet in overall MCC (Figure 2a). This result suggests that incorporating structural modality contributes to improved signal peptide prediction performance.

For the prediction of minor class signal peptides, we used Precision, Recall, and F1-score as evaluation metrics. Since earlier methods do not support the complete 6-class signal peptide classification and the evaluation of minor class signal peptides requires distinguishing between the 4 minor classes, the comparison in this study was limited to USPNet and SignalP6.0, both of which support 6-class classification. The results indicate that SaSPNet has a clear advantage in predicting minor class signal peptides. As shown in Figure 2b, SaSPNet improves Recall and F1-score by more than 10% compared to USPNet, with a significant increase in Precision as well. This strongly demonstrates that incorporating structural modality effectively enhances the model's ability to identify minor class signal peptides.

To further analyze SaSPNet's ability to recognize specific minor class signal peptides, we applied the One-vs-Rest strategy to compute the binary MCC for three minor class signal peptides, treating each signal peptide class as the positive class and combining the other five classes as the negative class for evaluation. As shown in Figure 2c, SaSPNet significantly outperforms other models in the Tat/SPII class, while also maintaining a leading position in the other two minor classes. This further validates the role of the structural modality in enhancing the prediction performance for minor class signal peptides.

3.2. Ablation Study

To validate the effectiveness of each encoder module in SaSPNet, particularly the contribution of the structural modality to the overall model performance, we conducted systematic ablation experiments under the same training conditions. We primarily focused on three encoders: the sequence encoder (Seq), the structure encoder (Struct), and the protein language model (PLM). Additionally, to investigate the design strategy of the structure encoder, we examined two variations: Struct(seq-init), where the sequence encoder features are used as the initial node features for the GCN, and Struct(embed-init), where the output of the embedding layer is used as the initial node features for the GCN.

As shown in Figure 2d, the complete SaSPNet model achieves the best performance across all evaluation metrics. After removing or modifying any encoder module, the model's performance declines to varying extents, demonstrating the rationality of the overall design of our model.

To investigate whether the sequence encoder truly plays an independent role, we first analyzed its importance. By comparing SaSPNet(full) with PLM+Struct(seq-init) and PLM+Struct(embed-init), we found that removing the sequence encoder features led to a performance drop, with a 1–2% decrease in MCC and a 3–7% decrease in the minor-class metrics. This indicates that the sequence encoder plays a crucial role in the model's representation.

Furthermore, the contribution of the structure encoder was validated across multiple experiments. The results of comparing SaSPNet(full) with Seq+PLM, as well as Seq+Struct with Seq-only, show that removing the structure encoder significantly degraded performance, with a general decrease of more than 5% in the minor-class metrics. This demonstrates that structural features enhance the representation of signal peptides. Although Struct-only performed worse compared to other variants, it still achieved acceptable results in both overall and minor-class tasks, indicating that the structure encoder is indeed capable of learning the representation of signal peptides. Additionally, our experimental analysis led to the identification of the optimal configuration for the GCN initial node features when using the protein language model: if the integrated features include sequence encoder features, the sequence encoder features should be used as the initial node features for GCN; if the integrated features do not include sequence encoder features, the output of the embedding layer should be used instead.

The protein language model also plays a crucial role in SaSPNet. Whether comparing SaSPNet(full) with Seq+Struct or Seq+PLM with Seq-only, the removal of PLM features resulted in a performance drop, indicating that PLM contributes significantly to the fusion features. Notably, PLM-only achieved an overall MCC of 0.90, but its performance on minor-class signal peptides was relatively poor. This suggests that while PLM contains rich evolutionary priors of proteins, it is still insufficient to fully capture the features of minor classes.

3.3. SP Cleavage-Site Prediction Performance

The determination of cleavage sites is a key step in signal peptide identification.^{38,39} To evaluate SaSPNet's performance on this task, we used Precision, Recall, and F1-score as metrics, calculated separately for all signal peptides and for minor-class signal peptides.

As shown in Figure 2e,f, SaSPNet achieves cleavage-site prediction performance comparable to USPNet for all signal peptides, indicating stable predictive capability for common classes. Meanwhile, for minor-class signal peptides, SaSPNet outperforms other methods across all three metrics, with an F1-score improvement of nearly 10% over USPNet, demonstrating that incorporating structural modality significantly enhances the model's ability to identify cleavage sites in minor classes.

3.4. Feature Space Visualization

To further analyze the feature representations learned by each encoder in SaSPNet, we performed dimensionality reduction and visualization using t-SNE⁴⁰ on the high-dimensional features extracted from different modules, including protein

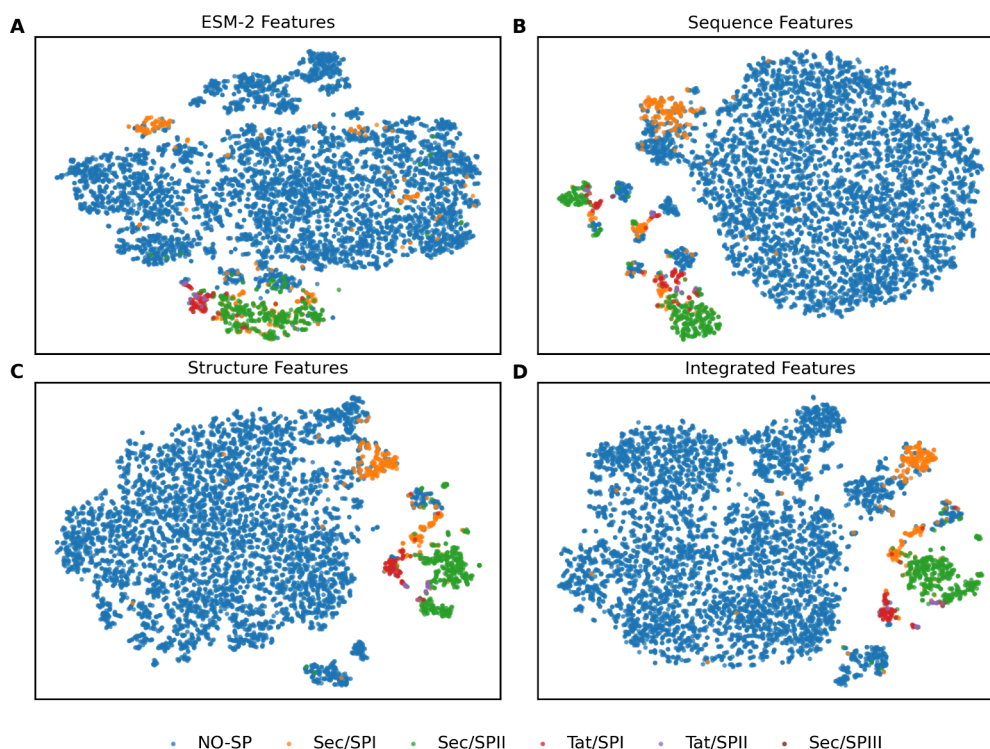


Figure 3. t-SNE visualization of multimodal feature representations learned by SaSPNet. (A) Protein language model features extracted using ESM-2. (B) Sequence encoder output features from the CNN–BiLSTM–attention module. (C) Structural features extracted by GCN. (D) Integrated features of SaSPNet, fusing sequence features, structural features, and protein language model features.

language model features, sequence encoder features, structural features, and integrated features, as shown in the four subplots of Figure 3. By comparing the distributions of features in different feature spaces, we can intuitively observe the contribution of each modality to the signal peptide representation.

From the visualization results, it can be observed that the protein language model (PLM) features extracted by ESM-2 show significant overlap between the minor-class signal peptide samples, indicating that PLM struggles to form sufficiently discriminative semantic representations for minor-class signal peptides. In contrast, the sequence encoder features show some improvement in class differentiation, particularly with a clear separation between NO-SP samples and other classes. However, confusion still exists between the minor classes.

In the feature space of the structure encoder, the clustering boundaries between classes are clearer, with a significant reduction in overlap among the minor-class signal peptides. This suggests that structural features capture the structural information needed to differentiate between different types of signal peptides. The clustering effect of the integrated features is similar to that of the structural features, further demonstrating the significant role of the structural modality in the multimodal integration.

3.5. Explainability Analysis of Structural Modality

To explain how the structural modality enhances the prediction performance of SaSPNet, we conducted an explainability analysis of the structure encoder. Specifically, we used GNNExplainer³⁴ on the benchmark test set to generate critical residue contact subgraphs for each sample, revealing the critical topological patterns learned by the GCN in the structural modality. We then computed the similarity between critical subgraphs of different samples using Jaccard

similarity and Cosine similarity, thereby obtaining the within-class and between-class critical subgraph similarity distributions for each signal peptide category.

As shown in Figure 4a,b, the within-class critical subgraph similarities for the minor classes (LIPO, TAT, TATLIPO; corresponding to Sec/SPII, Tat/SPI, and Tat/SPII in the transport pathway-based naming system) are significantly higher than those of the major classes (NO-SP, SP; corresponding to NO-SP and Sec/SPI), indicating that the structure encoder has learned more consistent and representative structural patterns in the minor-class signal peptides. Here, we adopt the sequence-based naming system (SP, LIPO, PILIN, TAT, TATLIPO, NO-SP) for consistency with the visualization in Figure 4, while these categories are in one-to-one correspondence with the transport pathway-based naming system used in other sections. This suggests that an important reason for the performance improvement of SaSPNet in minor-class prediction is that the structural modality captures topological features shared by minor-class samples, thereby enhancing within-class consistency and between-class separability in the representation space. To further verify the statistical significance of this result, we divided the similarities of the major and minor classes into two groups and performed a *t* test. The results show that for Jaccard similarity, $t = -24.678$, $p = 1.700 \times 10^{-131}$; for Cosine similarity, $t = -31.834$, $p = 2.748 \times 10^{-214}$. These results strongly support the conclusion that the critical subgraph similarities between minor-class samples is significantly higher than those of the major classes.

From a biological perspective, the critical residue contact subgraphs identified by GNNExplainer can be interpreted as structural motifs that are most influential for the model's predictions. These subgraphs capture key residue–residue

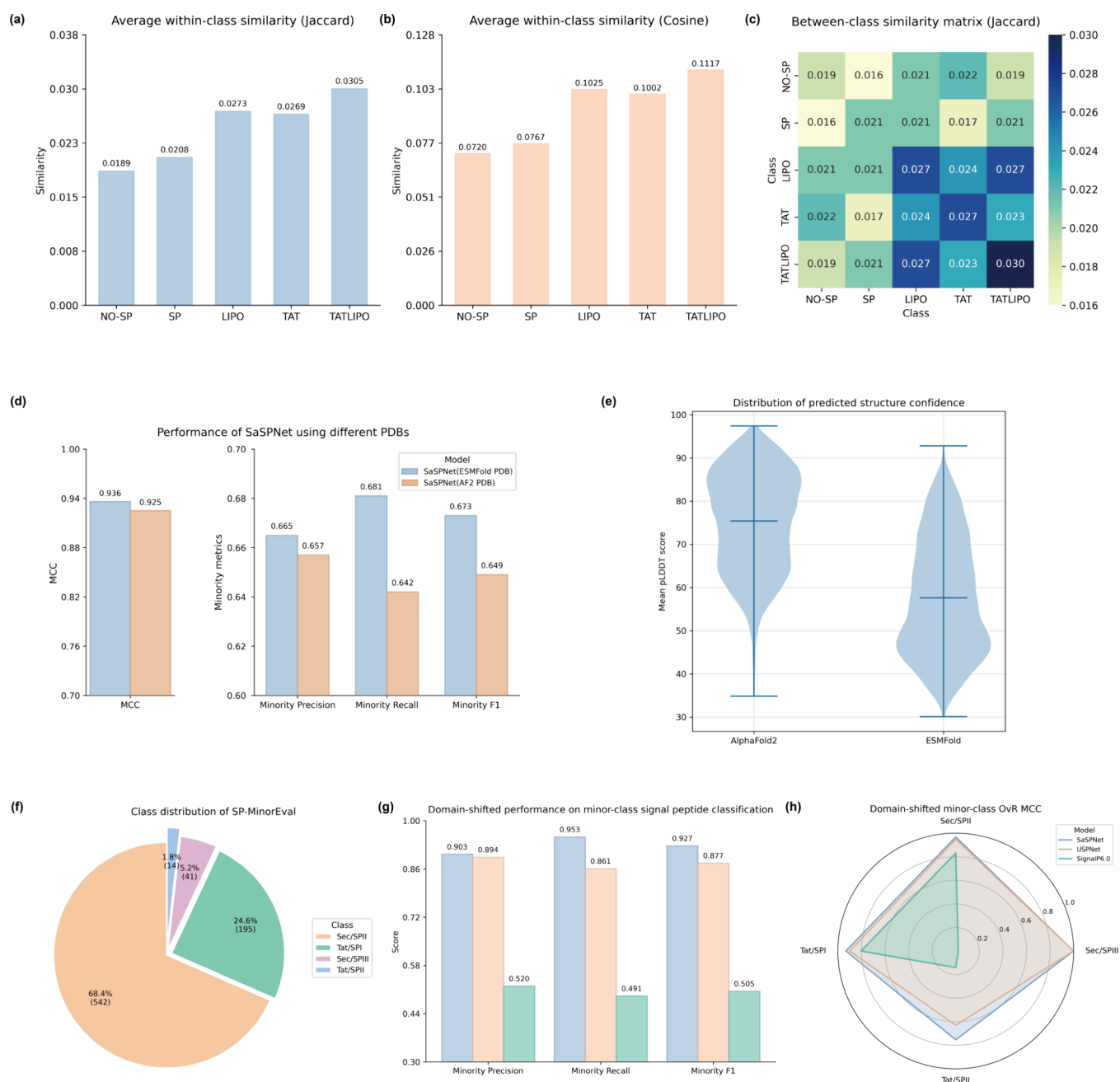


Figure 4. Explainability analysis, performance using AlphaFold2 PDBs, and domain-shifted generalization of SaSPNet. (a,b) Within-class subgraph similarities across SP classes. We use Jaccard similarity and Cosine similarity to measure the similarity of critical residue contact subgraphs. The signal peptide class labels follow the sequence-based naming system (SP, LIPO, TAT, TATLIPO, and NO-SP), which correspond one-to-one with the transport pathway-based categories (Sec/SPI, Sec/SPII, Tat/SPI, Tat/SPII, and NO-SP). (c) Heatmap of between-class subgraph similarities across SP classes. Similarities are computed using Jaccard similarity. (d) Performance comparison of SaSPNet trained with ESMFold PDBs and AlphaFold2 PDBs, including classification results on overall and minor-class SPs. (e) Violin plot comparing pLDDT scores of AlphaFold2 PDBs with those of ESMFold PDBs. (f) Composition of sample classes in SP-MinorEval. The data set contains 792 samples distributed across four minor-class signal peptide categories. (g,h) Performance of models on SP-MinorEval. Metrics include Precision, Recall, F1-score, and One-vs-Rest MCC computed for each minor-class.

interactions within the signal peptide structure, which may reflect underlying structural constraints associated with different signal peptide types. For example, signal peptides typically exhibit characteristic organization patterns, and specific types such as LIPO and TAT are known to possess distinct sequence and structural features related to their transport mechanisms. The higher within-class similarity observed in minor classes suggests that these types may share more conserved structural

interaction patterns, which are effectively captured by the structure encoder.

In addition, the between-class critical subgraph similarity matrix for signal peptide classes (Figure 4c) further indicates that the within-class similarities for the minor classes is generally higher than the between-class similarities. It is noteworthy that the within-class similarity for NO-SP is relatively low, as NO-SP is not a single functional type but

includes various nonsignal peptide fragments, such as transmembrane regions and signal anchors, which have highly diverse structural patterns. Furthermore, the overall similarity values are relatively low, which is reasonable because different signal peptide samples exhibit inherent differences at the three-dimensional structural level. However, despite these individual structural differences, the model is still able to extract shared topological feature patterns within the class in the structural modality, thereby effectively improving the prediction performance for minor-class signal peptides.

3.6. Performance Using AlphaFold2-Generated PDBs

In the model training and evaluation phases, the three-dimensional structural data of signal peptides used by SaSPNet were generated by ESMFold²⁶ To verify the adaptability of SaSPNet to different structural data sources, we further retrained and evaluated the model using PDB structures generated by AlphaFold2,²⁵ and compared the results with those obtained using the ESMFold version.

As shown in Figure 4d, SaSPNet using AlphaFold2 PDB performs slightly worse than the version using ESMFold PDB, indicating that three-dimensional structural data from different sources can indeed affect the model's predictive performance. However, regardless of whether the structural data is generated by ESMFold or AlphaFold2, SaSPNet's performance is consistently superior to models that do not include the structural modality (see Figure 2a,b). This demonstrates that the introduction of the structural modality can consistently enhance the model's overall performance and indicates that SaSPNet has good adaptability to signal peptide structures generated by mainstream structural prediction models.

To further explore the relationship between the quality of structural data and model performance, we evaluated the average quality of ESMFold PDB and AlphaFold2 PDB using pLDDT. pLDDT reflects the confidence in the structural prediction for each residue, with higher values indicating more reliable structural predictions for that position. As shown in Figure 4e, the PDB generated by AlphaFold2 has a higher average pLDDT value overall, indicating better prediction quality. However, this result presents a somewhat counter-intuitive relationship with model performance. Despite AlphaFold2's higher structural quality, SaSPNet using ESMFold PDB achieved better performance. We believe this may be related to the structural characteristics of signal peptides themselves. AlphaFold2's predictions are more closely aligned with the folding of full-length proteins, while signal peptides are generally shorter and more flexible⁴¹ with structural features that differ from those of the complete protein fold. As a result, the more precise structure predicted by AlphaFold2 may deviate from the actual conformation of the signal peptide. In contrast, while the structures generated by ESMFold are coarser, they may better align with the local spatial features of the signal peptide.

This finding indicates that higher-quality AlphaFold2 PDB does not necessarily lead to better classification performance, but it can still provide gains for the recognition of minor-class signal peptides. It also shows that SaSPNet does not rely on the most precise structural prediction models and can effectively utilize the information provided by lightweight structure prediction tools. Considering that ESMFold is significantly faster than AlphaFold2 in inference, and that SaSPNet already achieves strong performance using ESMFold PDB, this enables SaSPNet to maintain high efficiency and

scalability for applications on large-scale data sets, such as metagenomic signal peptide mining.

3.7. Generalization on the Independent Test Set

SaSPNet was trained on the USPNet training set, and since the benchmark test set belongs to the same data domain as the training set, strong performance on the benchmark test set is expected. To further evaluate SaSPNet's domain-shifted generalization ability for minor-class signal peptide prediction, we constructed an independent test set, SP-MinorEval, which contains only four minor-class signal peptides: Sec/SPII (LIPO), Sec/SPIII (PILIN), Tat/SPI (TAT), and Tat/SPII (TATLIPO), with the specific composition shown in Figure 4f.

On SP-MinorEval, we evaluated the signal peptide classification performance of SaSPNet, USPNet, and SignalP6.0, using the same metrics as the previous minor-class performance evaluation, including Precision, Recall, F1-score, and One-vs-Rest MCC for each class. As shown in Figure 4g,h, SaSPNet still achieves the best performance on the independent test set across all evaluated minor classes. Notably, it demonstrates strong predictive capability for Sec/SPIII (PILIN), a class that is newly introduced in this data set and not explicitly evaluated in the original benchmark. This result highlights the robustness of SaSPNet under domain shift, as SP-MinorEval differs from the training data in both class composition and distribution. By focusing exclusively on minor-class signal peptides, this data set provides a more challenging and targeted evaluation of the model's ability to generalize beyond the training domain without being influenced by dominant classes.

The consistently superior performance of SaSPNet suggests that the incorporation of structural modality enables the model to capture more transferable and class-relevant structural patterns, particularly for minor classes. These patterns appear to be less sensitive to data set-specific variations, thereby supporting improved generalization in cross-domain scenarios.

4. DISCUSSIONS

The signal peptide prediction task is commonly faced with extreme data imbalance, with the prediction performance for minor-class signal peptides lagging behind that of the major-class. To address this issue, we propose the multimodal deep learning model SaSPNet, which introduces a structural modality on top of the traditional sequence modality. By employing a Graph Convolutional Network (GCN) to model the 3D structure of signal peptides, SaSPNet captures discriminative information at the structural level. Experimental results demonstrate that SaSPNet achieves significant performance improvements in both minor-class signal peptide classification and cleavage-site prediction tasks, proving the effectiveness of the structural modality in compensating for the insufficient representation of minor classes. Furthermore, through feature visualization and interpretability analysis, we further reveal the role of the structural modality in the model's decision-making process, showing that the GCN is capable of capturing more consistent structural patterns within minor-class signal peptides.

Although SaSPNet demonstrates superior predictive performance, there is still room for improvement. First, SaSPNet relies on the 3D structural data of signal peptides during both the training and inference stages. These structures typically need to be generated using protein structure prediction methods, which, even with faster inference tools like ESMFold,

add extra computational overhead. This results in data preprocessing times that are longer than those of pure sequence-based models. Second, the structure encoder is implemented based on GCN, which incurs higher computational complexity and memory consumption compared to the sequence encoder, thereby increasing the overall computational cost of the model. Moreover, the predictive performance of the structural modality is influenced by the quality of the input structures; if the structural predictions contain errors, this may lead to performance fluctuations.

Future research could explore incorporating structural data during the model training phase to learn the feature representations of the structural modality, while only using sequence information for prediction during the inference phase. This design would not only significantly reduce data preprocessing time and computational load during inference but also reduce the model's dependence on the quality of structural predictions. In this way, the advantages of the structural modality can still be leveraged while ensuring efficiency. Through such improvements, we can develop signal peptide prediction tools that balance both high performance and high efficiency, providing more reliable support for signal peptide research and functional protein discovery.

■ ASSOCIATED CONTENT

Data Availability Statement

The data used in this study and the source code are openly available for those interested at <https://github.com/YanngYT/SaSPNet>.

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Author Contributions

Y.Y. designed the study, performed the experiments, analyzed the results, and wrote the manuscript. S.X. contributed to literature review and assisted with experimental implementation. S.C. was responsible for figure preparation and visualization. J.Q. contributed to data processing and organization. H.Z. assisted with manuscript writing and revision. Z.Y. provided support in experimental validation and result analysis. L.W. supervised the project, provided conceptual guidance,

and contributed to manuscript revision. All authors reviewed and approved the final manuscript.

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Notes

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